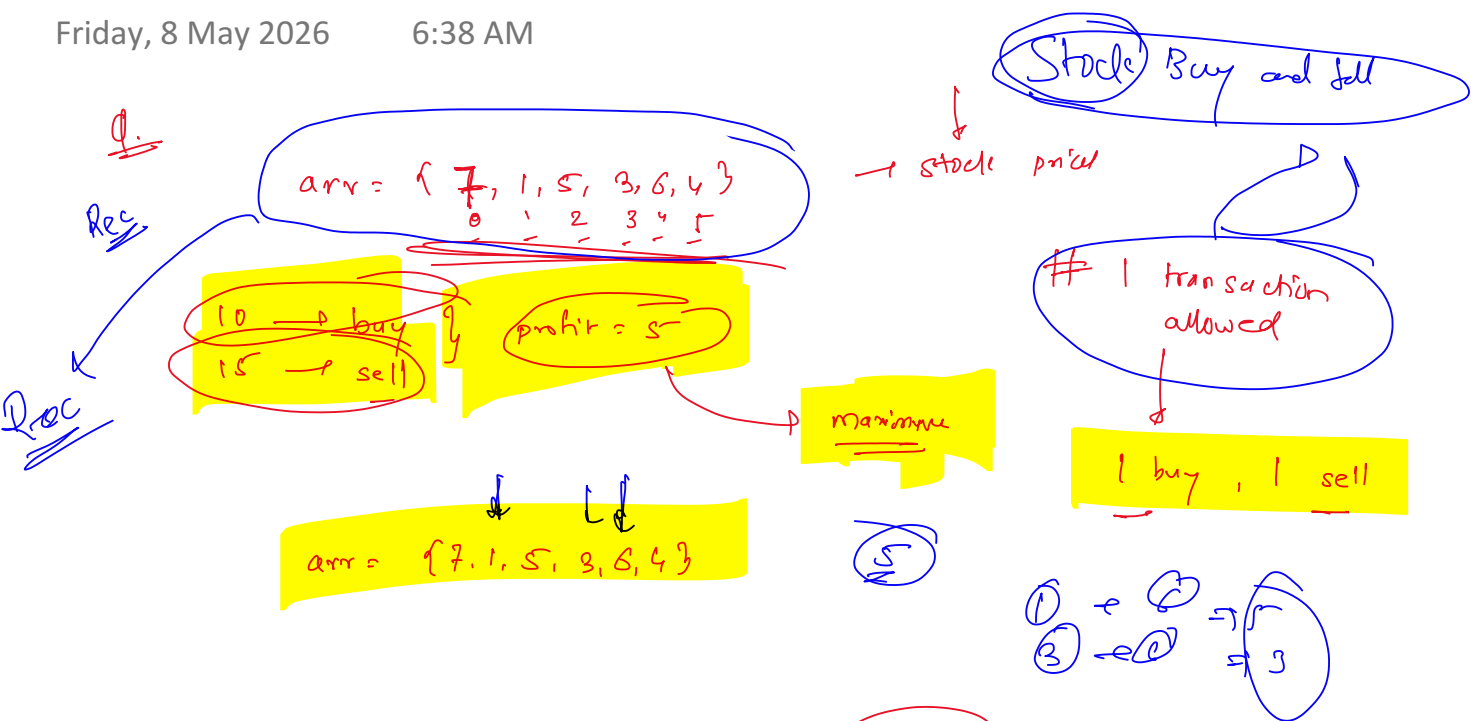


Arrays - IX

Friday, 8 May 2026 6:38 AM

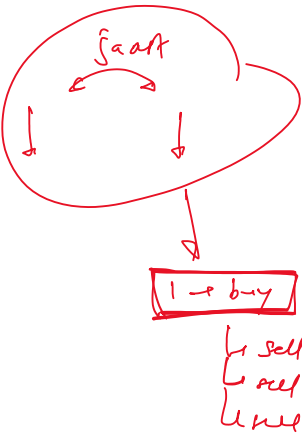


Approach 1 - 2 for loop

{ 7, 1, 5, 3, 6, 4 }



{ 7, 1, 5, 3, 6, 4 }



Code-

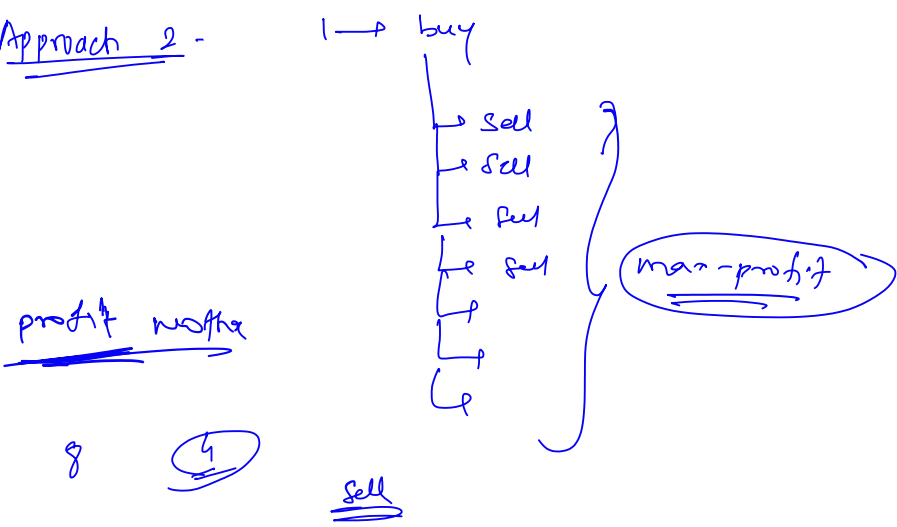
{ 7, 6, 5, 4, 3 }

```
int max=0;

for (int i=0; i<arr.length; i++) {
    for (int j=i+1; j<arr.length; j++) {
        int profit = arr[j]-arr[i];
        max = Math.max(max, profit);
    }
}
```

// TC - O(n²)
// SC - O(1)

Approach 2 -



{ 7, 1, 5, 3, 6, 4 }

profit maximize ⇒ 1) Buy low →
2) Sell High

Dry Run -

arr = { 7, 1, 5, 3, 6, 4 }

stock	price	low	profit	max-profit
7		7	0	0
1		1	0	0
5		1	4 (5-1)	4
3		1	2 (3-1)	4
6		1	5 (6-1)	5
4		1	3 (4-1)	5

ans = 5

Code-

```
int lowPrice = Integer.MAX_VALUE;

int max = 0;

for (int i=0; i<arr.length; i++) {
    lowPrice = Math.min(lowPrice, arr[i]);
    int profit = arr[i] - lowPrice;
    max = Math.max(max, profit);
}
```

// TC - O(n)
// SC - O(1)